

Final methodology for extraction

1. Conduct a pilot study on the 3 specifications to extract the primary units by two human annotators, and the agreement is calculated between the two annotations. A 1 is given if an annotator uses the primary unit to extract recommendations, and a 0 is given if an annotator does not use the primary unit to extract recommendations.

Primary unit: A sentence, a row of a table, a bullet in a bullet list, or the caption of a figure.

Note: The caption of a figure is considered sufficient as a primary unit instead of separating the figure itself into individual units, since figures have empirically been a graphical representation of already existing information in text format, represented in a graphical format. So the loss of information of the details of the figure should not have a drastic affect on the final recommendations

2. On the basis of the pilot study, extract the individual recommendations from the entirety of the 3 specifications. Given a sufficiently high agreement between notebook LM and human annotators, this part can be automated using notebook LM.

Questions:

1.1 Do we need to repeat calculating the agreement for the entire specifications as we did in the pilot study? Or can the generalisation of the pilot study (extracting primary units from 10-15% of the entire specifications) to extracting individual recommendations be made?

1.2 Does the agreement in extracting *primary units* directly translate to authenticity for the extracted *recommendations*?

Question 2: A hybrid LLM-based approach seems a very arbitrary approach to finalising the set of recommendations. A purely LLM-based approach, based on the pilot study's success or a completely manual approach with two annotators seems more robust. (More on this below)

Approach for extractions:

1. Manual Annotators

- Annotators A, B, and C extract recommendations from the entire specifications in a manner that two annotators extract from each document.
- Cohen's Kappa is then computed for both annotators of each specification to be reported in our methodology (for robustness of the paper).
- Afterwards, both the annotators sit together to resolve discrepancies, resulting in a consolidated, uncontested set of recommendations (much like how the research community approaches such nuances) [1].

2. NotebookLM + Manual Annotators

- Annotators employ a semi-LLM approach here (i.e., use LLMs and verify via manual inspection whether the extractions are fine).
- . How is Cohen's Kappa calculated here? Is the approach robust? Might raise concerns from reviewers? Also, didn't find literature doing such executions in their studies.

3. NotebookLM/Any other LLM

- We employ LMs to generate all the recommendations from the specifications.
- How do we compute the Cohen's Kappa here? What's the use of keeping the Primary Source Unit in such a case then?

[1] Tsinganos, N.; Mavridis, I. Building and Evaluating an Annotated Corpus for Automated Recognition of Chat-Based Social Engineering Attacks. Appl. Sci. 2021, 11, 10871. <https://doi.org/10.3390/app112210871>